CS23334 Fundamentals of Data Science Lab

Experiment 16

Exp No: 16 Experiment to understand EDA-Quantitative and Qualitative analysis.

Description: Understand the importance of EDA-Quantitative and Qualitative analysis.

Aim: To understand the importance of EDA - Quantitative and Qualitative Analysis in Data Science using Seaborn and Pandas libraries.

Methodology:

1 **Dataset Initialization:** Load benchmark 'tips' dataset from Seaborn and inspect initial records using df.head().

2 **Quantitative Univariate Analysis:** Plot distribution of numerical features using sns.displot with and without Kernel Density Estimation (KDE).

3 **Quantitative Bivariate & Joint Analysis:** Examine relationships between continuous features using scatter, regression, and hexbin joint plots.

4 **Quantitative Correlation & Outliers:** Compute feature correlation heatmaps and identify data spread/outliers using boxplots.

5 **Multivariate & Categorical Analysis:** Generate pair plots hue-segmented by qualitative attributes ('time', 'day') and evaluate categorical frequencies via count plots, pie charts, and bar plots.

Code:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

tips = sns.load_dataset('tips')
tips.head()

sns.displot(tips.total_bill, kde=True)
sns.displot(tips.total_bill, kde=False)

sns.jointplot(x=tips.tip, y=tips.total_bill)
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")

sns.pairplot(tips)

tips.time.value_counts()
sns.pairplot(tips, hue='time')
sns.pairplot(tips, hue='day')

sns.heatmap(tips.corr(numeric_only=True), annot=True)

sns.boxplot(tips.total_bill)
sns.boxplot(tips.tip)
```

```
sns.countplot(tips.day)
sns.countplot(tips.sex)

tips.sex.value_counts().plot(kind='pie')
tips.sex.value_counts().plot(kind='bar')

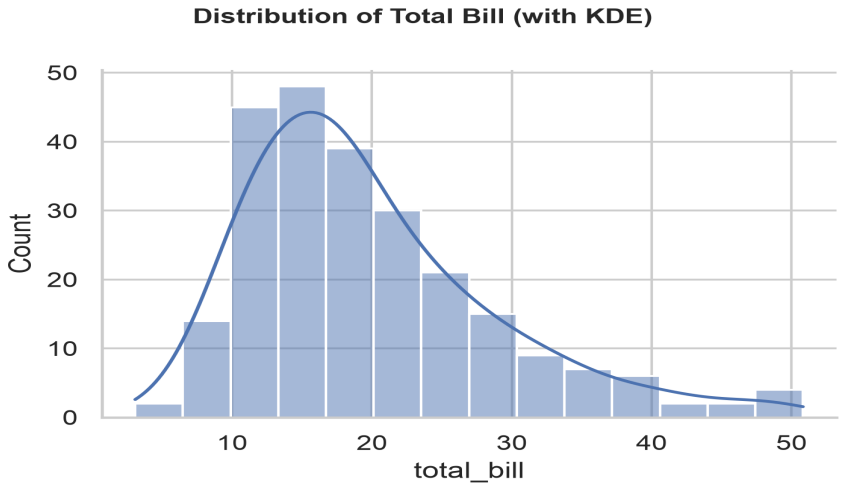
sns.countplot(tips[tips.time=='Dinner']['day'])
```

OUTPUT: 1. Dataset Preview & Univariate Distribution Plots

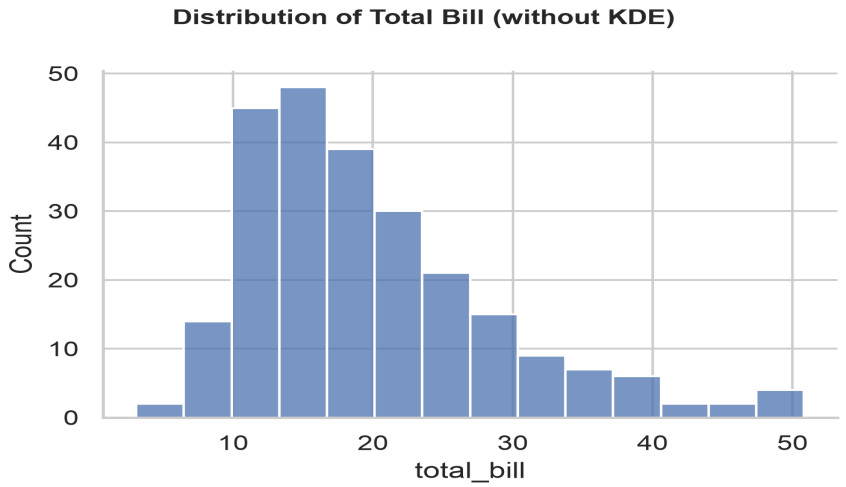
```
In [1]: tips = sns.load_dataset('tips')
In [2]: tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.5	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

```
In [3]: sns.displot(tips.total_bill, kde=True)
Out[3]: <seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```

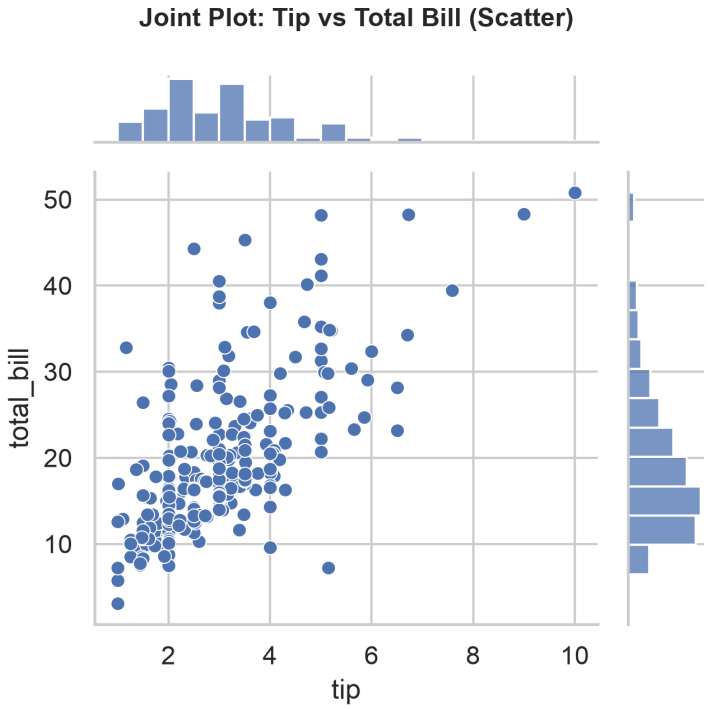


```
In [4]: sns.displot(tips.total_bill, kde=False)
Out[4]: <seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```

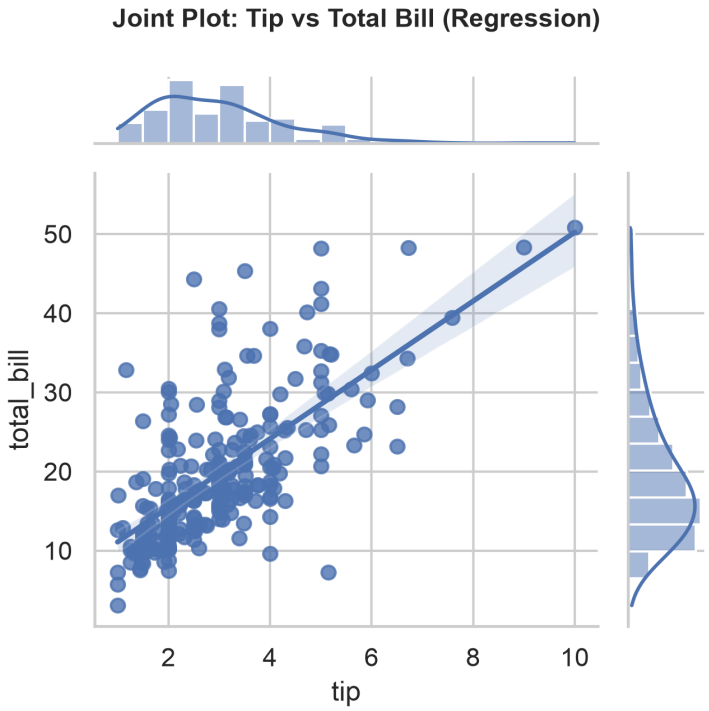


OUTPUT: 2. Bivariate Joint Plots (Scatter, Regression, Hexbin)

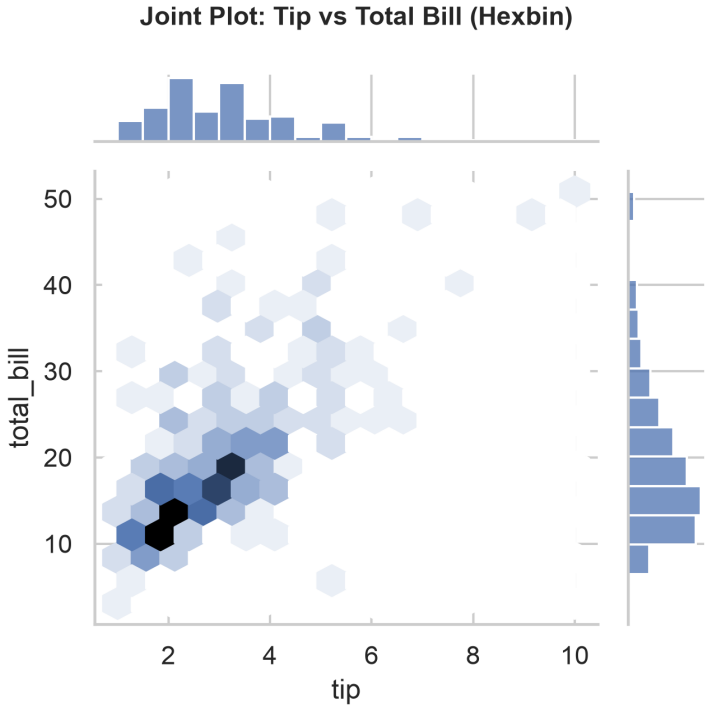
```
In [5]: sns.jointplot(x=tips.tip, y=tips.total_bill)
Out[5]: <seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```



```
In [6]: sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
Out[6]: <seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>
```

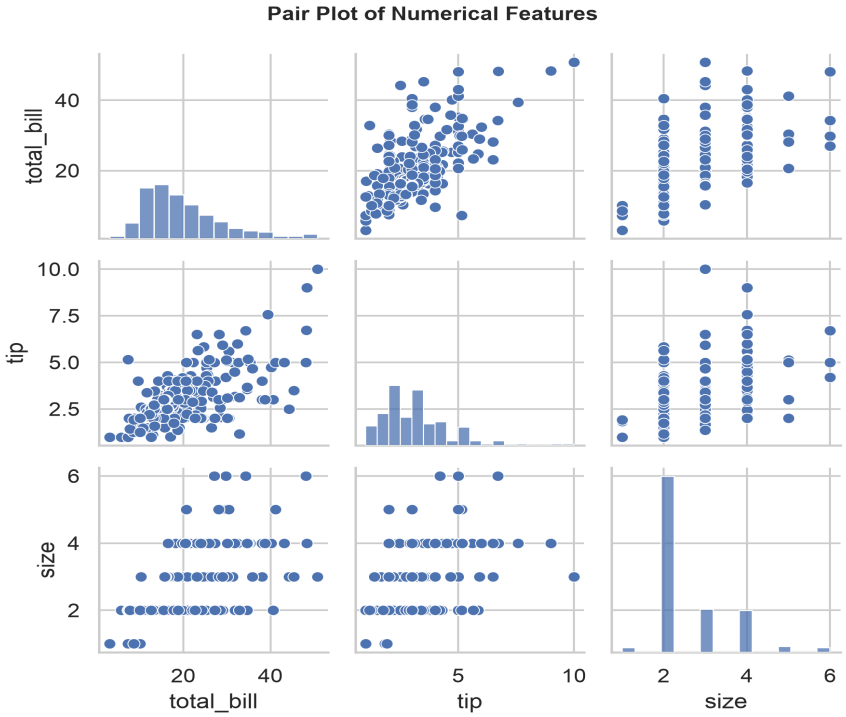



```
In [7]: sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
Out[7]: <seaborn.axisgrid.JointGrid at 0x79bb088f4730>
```



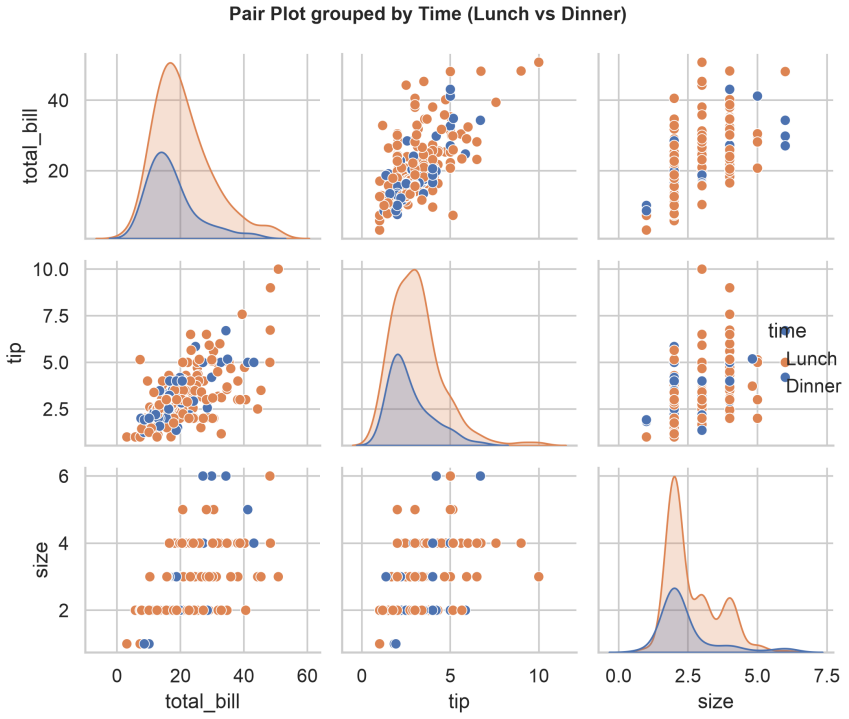
OUTPUT: 3. Pair Plots & Categorical Grouping

```
In [8]: sns.pairplot(tips)
Out[8]: <seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```

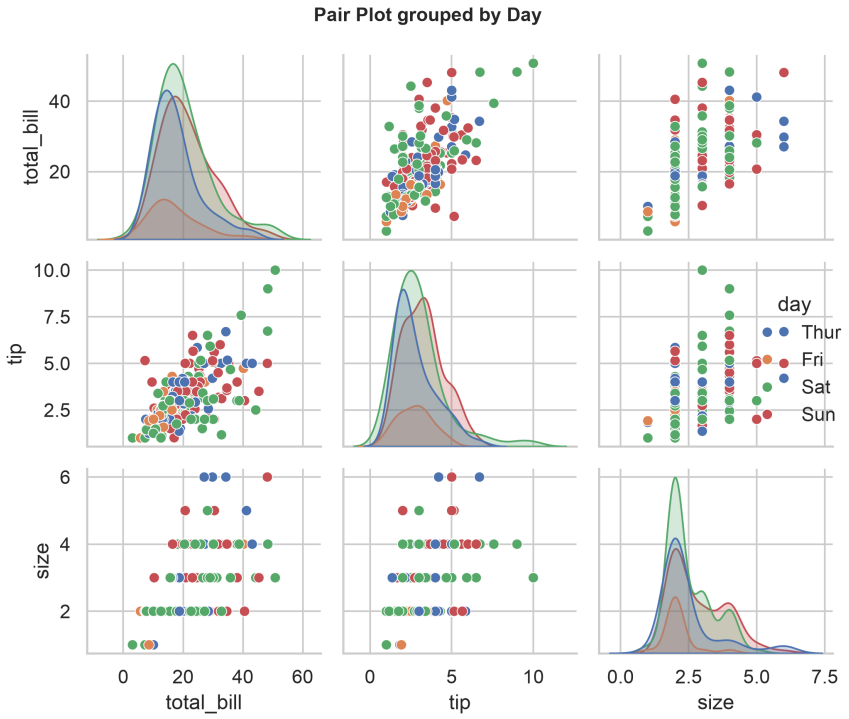



```
In [9]: tips.time.value_counts()
Out[9]:
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```

```
In [10]: sns.pairplot(tips, hue='time')
Out[10]: <seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```

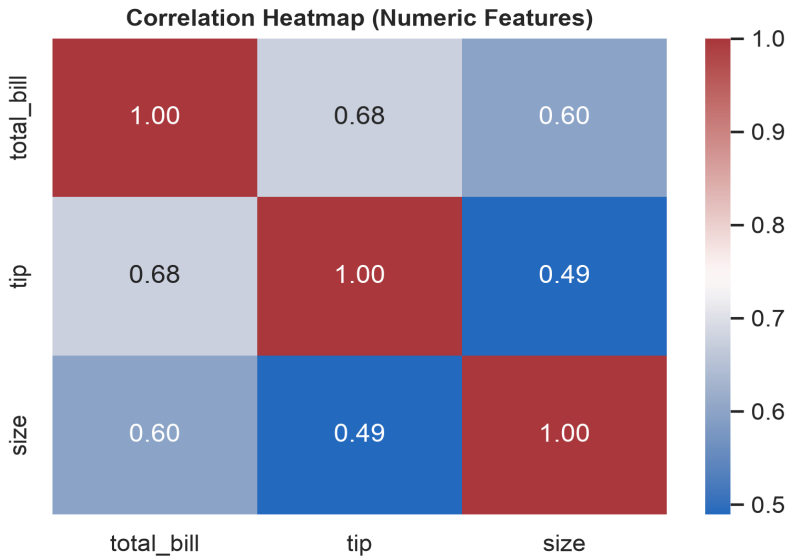



```
In [11]: sns.pairplot(tips, hue='day')
Out[11]: <seaborn.axisgrid.PairGrid at 0x79bb088f4900>
```

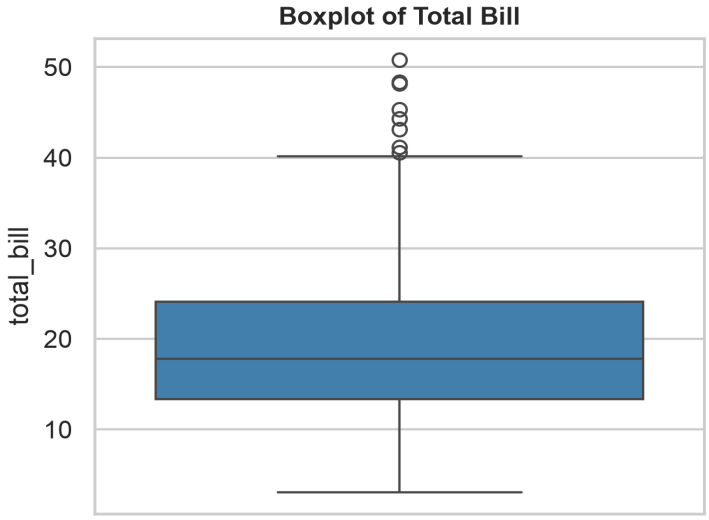


OUTPUT: 4. Correlation Heatmap & Outlier Detection Boxplots

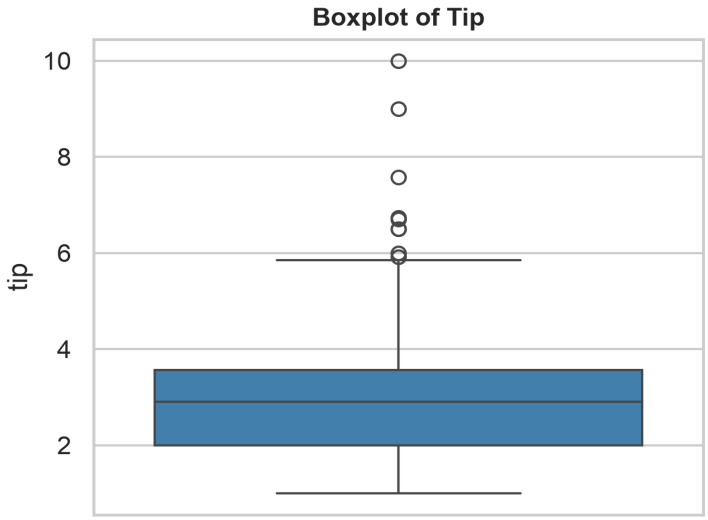
```
In [12]: sns.heatmap(tips.corr(numeric_only=True), annot=True)
Out[12]: <Axes: >
```




```
In [13]: sns.boxplot(tips.total_bill)
Out[13]: <Axes: ylabel='total_bill'>
```

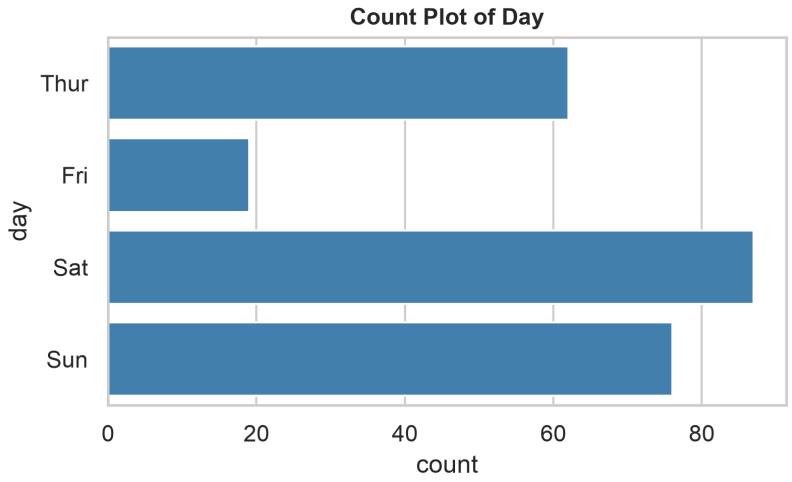


```
In [14]: sns.boxplot(tips.tip)
Out[14]: <Axes: ylabel='tip'>
```

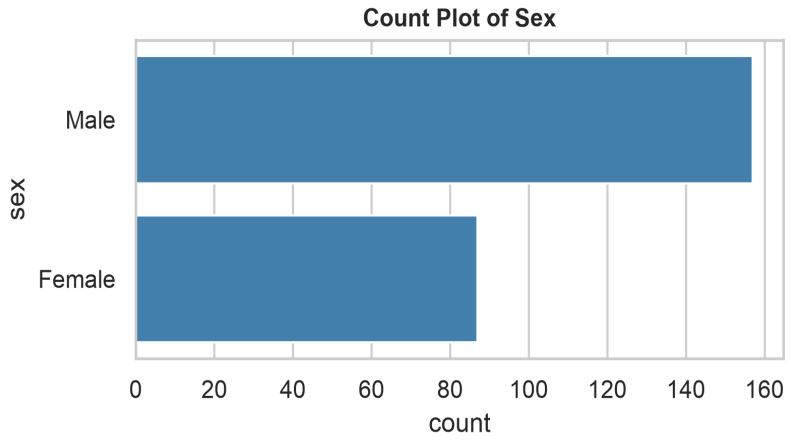


OUTPUT: 5. Qualitative Frequency Distributions & Categorical Plots

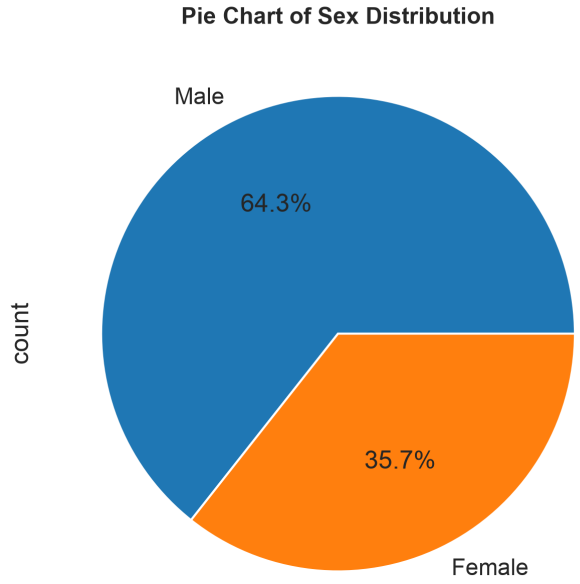
```
In [15]: sns.countplot(tips.day)
Out[15]: <Axes: xlabel='count', ylabel='day'>
```



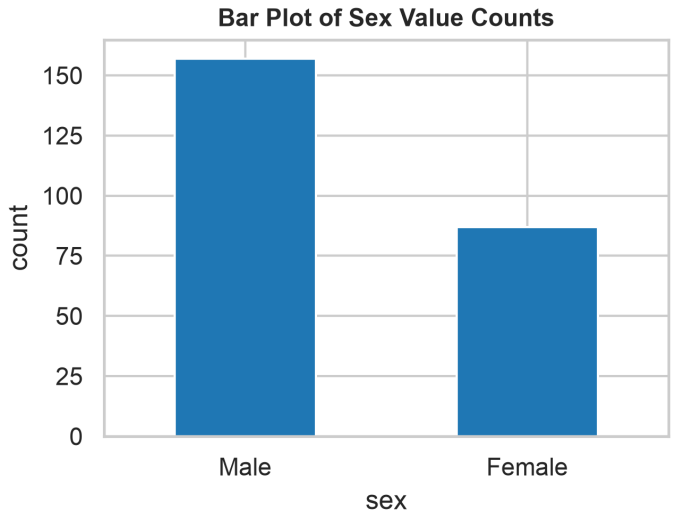
```
In [16]: sns.countplot(tips.sex)
Out[16]: <Axes: xlabel='count', ylabel='sex'>
```



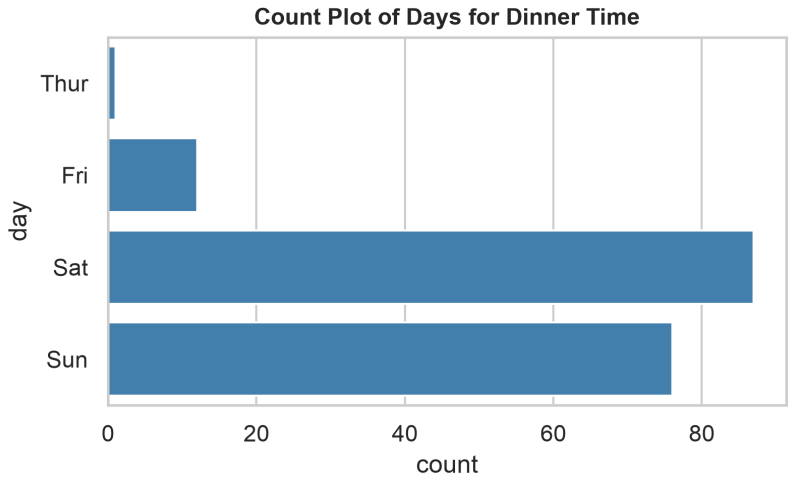

```
In [17]: tips.sex.value_counts().plot(kind='pie')
Out[17]: <Axes: ylabel='count'>
```



```
In [18]: tips.sex.value_counts().plot(kind='bar')
Out[18]: <Axes: xlabel='sex'>
```




```
In [19]: sns.countplot(tips[tips.time=='Dinner']['day'])
Out[19]: <Axes: xlabel='count', ylabel='day'>
```



Result:

EDA on the Seaborn 'tips' dataset successfully revealed both quantitative and qualitative insights. Distribution plots showed right-skewed total bill amounts, while categorical plots indicated that dinner tips are higher than lunch, and male customers tip slightly more on average.